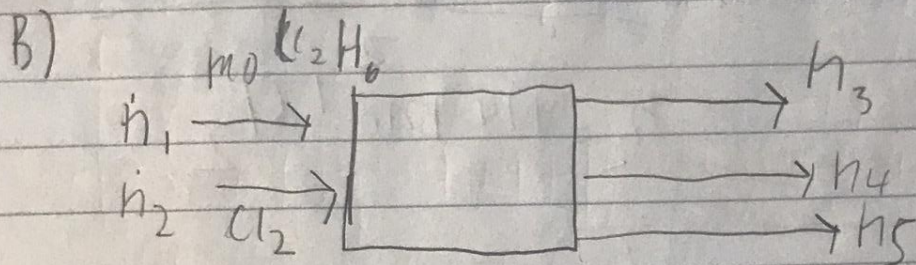


Jacob Tyrawa

64
72

a) the reactor would be low conversion of ethane, ✓



$$\begin{array}{r} 4a) 5 \\ + 4b) 2 \\ \hline 7 \end{array}$$

$$\begin{array}{r} 5a) 1 \\ b) 0 \\ c) 0 \end{array}$$

$$\begin{array}{r} +d) 4 \\ \hline 5 \end{array}$$

c) Atomic Balance

$$6n_1 = n_3 + 5n_4 + 6n_5 + 4n_6$$

$$2n_1 = 2n_4 + 2n_6 + 2n_5$$

$$2n_2 = n_4 + 2n_6 + n_3$$

Cl balance

$$2n_1 = 2n_4 + 2n_6 + 2n_5$$

$$2n_1 = 2 \cdot 100 + 2 \cdot 7.143 + 2(1 - 0.15)n_1$$

$$n_1 = 714 \text{ mol } C_2H_6$$

H balance

$$6 \cdot 714 = n_3 + 5(100) + 6(607.14) + 4(7.143)$$

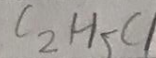
$$n_3 = 113 \text{ mol } HCl$$

2 und entsprechend

$$n_6 = 7.143$$

$$(1 - 0.15)n_1$$

$$100 \text{ mol } n_4$$



C1

$$2h_2 = h_4 + 2h_6 + h_3$$
$$h_2 = \frac{100 + 2(7.143) + h_3}{2}$$

$$h_2 = 114 \text{ mol Cl}_2$$

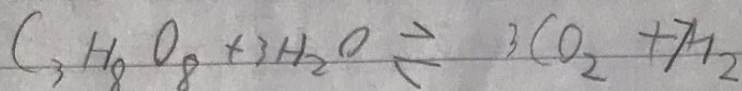
$$\text{Feed: } \frac{114}{714} = .1596$$

$$\frac{100}{1.14} = 87.7\%$$

D) The process was giving a higher wt% so this would mean a more chlorinated product

64/74

4.40

Glycerin

IN = out + CONSUM

$$\dot{h}_1 - \dot{q}_1 = h_{1,out}$$

$$\frac{(150)}{92} = h_{1,out} - \dot{q}_1$$

$$h_1 = .2304$$

Water

$$5 \left(\frac{150}{92} \right) - 3\dot{q}_1 = h_{1,out}$$

$$h_{1,out} = 3.95$$

CO₂

$$0 = h_3 + 3\dot{q}_1$$

$$h_3 = 4.2$$

H₂

$$0 = h_4 + 7\dot{q}_1$$

$$h_4 = 7.8$$

$$.54 = \frac{7\dot{q}_1}{(1.63 - \dot{q}_1) + (7\dot{q}_1) + 3\dot{q}_1 + (8.14 - \dot{q}_1)}$$

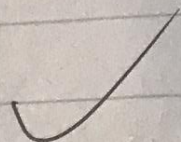
$$.54 = \frac{7\dot{q}_1}{9.77 + 6\dot{q}_1}$$

$$5.27 + 3.24\dot{q}_1 = 7\dot{q}_1$$

$$\dot{q}_1 = 1.4$$

$$K_{ut} = \frac{1.63 - .23}{1.63}$$

$$85.9\%$$



$$x_{CO_2} = \frac{4.2}{18.7} = .231 \quad y_{H_2} = .54$$

$$x_{C_3H_8O_3} = .0127 \quad y_{H_2O} = .217$$

$$K_a = \frac{(.231)^3 (.540)^7}{(.0127)(.217)^3} \cdot (1.2)^6 = 3.8 \text{ atm}$$

Not in eq ✓

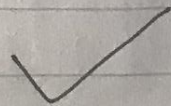
using numerical methods our new $\xi_1 = 1.57$

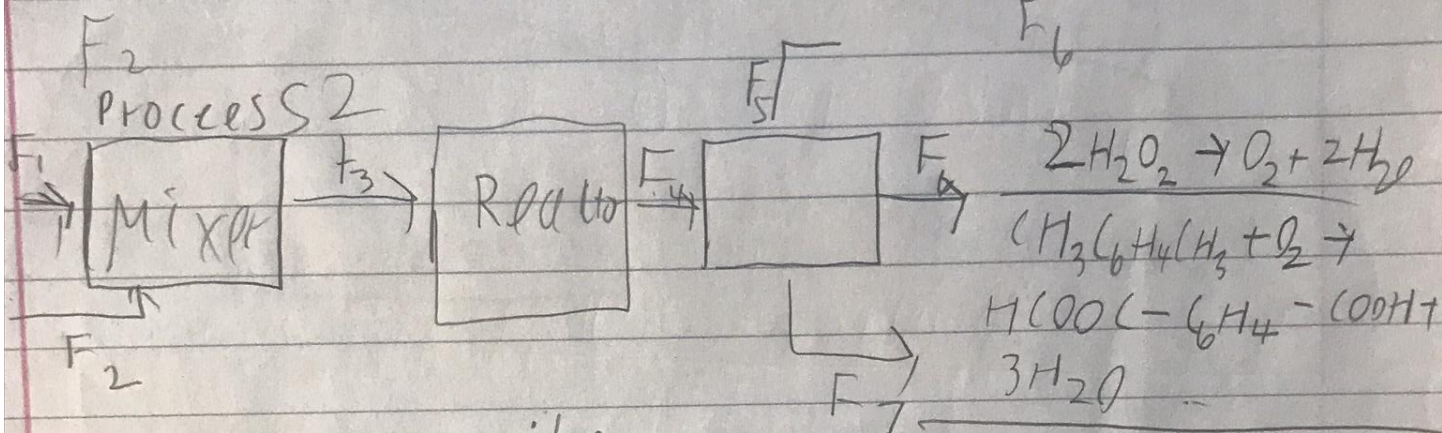
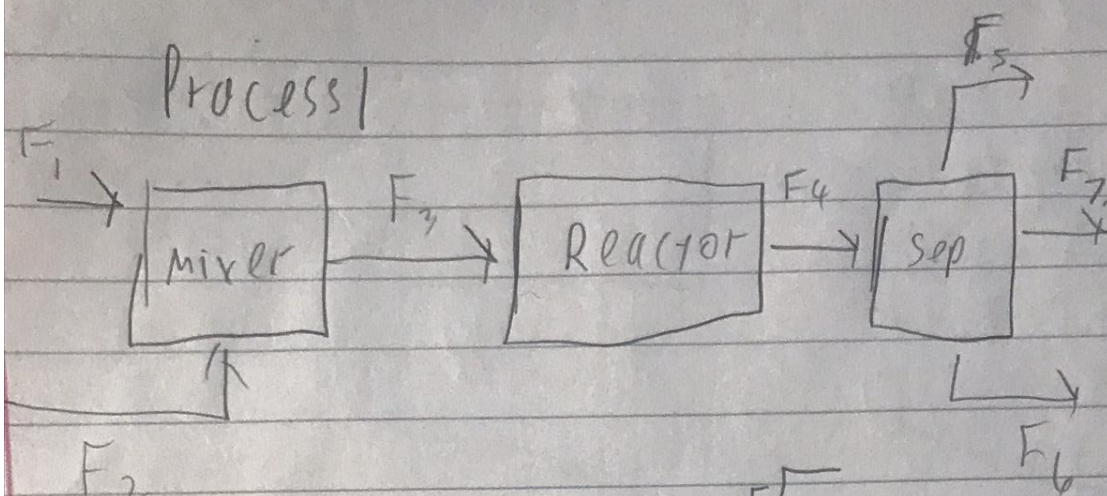
$$y_{H_2} = \frac{\xi_1}{9.77 + 6\xi_1} = \frac{7(1.57)}{9.77 + 6 \cdot 1.57} = .57$$

$$\beta_c = \frac{1.63 - .06}{1.63} = .96$$

$$n_{C_3H_8O_3} = 1.63 - \xi_1 \Rightarrow .06$$

we would improve the constant with pressure and Feed Ratio.

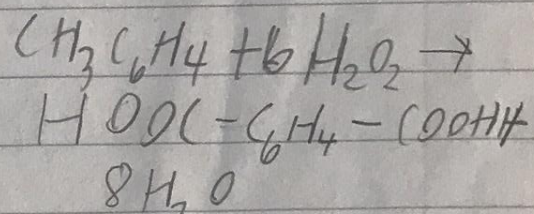




conversion yield

Process 2

$$\frac{166}{166 + 6.3481} = 54\% \quad \checkmark$$



correct chem

$$\frac{166}{166 + 3.32} = 82\% \quad \leftarrow$$

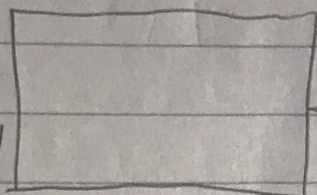
Calculation
Conclusion

~~185~~
3

④

4.49

100 CH₃OH
100 NH₃



9.4% CH₃OH

2.0% NH₃

18.6% CH₃NH₂

10.7% (CH₃)₂NH

41.1% H₂O

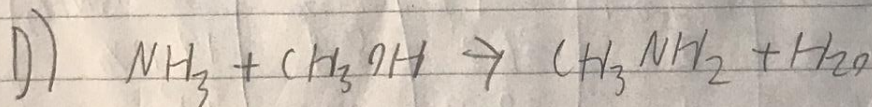
N: 100 =

$$100 + 100 + \xi_1 (-1 - 1 + 1 + 1) + \xi_2 (-1 - 1 + 1) = 200$$

a) $\frac{100 - (.094 \cdot 200)}{100} = .81$ CH₃OH ← yield ✓

b) $\frac{.186 \cdot 200}{100} = .37$ CH₃NH₂ ← yield

c) Selectivity CH₃NH₂ $\frac{.186 \cdot 200}{100} = .37$ ✓



$$N_{A, \text{out}} = N_{A, \text{in}} + \xi_{\text{NH}_3}$$

$$40 = \xi_1 = 60$$

✓

$$(CH_3)_2NH_4 = .109 \cdot 200 = 21.8$$

$$\delta_2 = 21.8$$

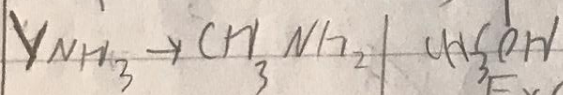
$$K_{a1} = \frac{.186 \cdot 1.411}{.2 \cdot .1094} = 4.066 \approx 4$$

$$K_{b1} = 2.5 = \frac{.186 \cdot .1094}{.2 \cdot .141} = 2.56 \approx 2$$

$$K_A = \frac{(\delta_1 - \delta_2) \cdot .186 \cdot (.1094)}{(n_{in} - \delta_1 - \delta_2) (n_{NH_3} - \delta_1 - \delta_2)} = 4$$

$$K_{b1} = 2.5 = \frac{\delta_2 (\delta_1 + \delta_2)}{(\delta_1 - \delta_2)}$$

n_{NH_4}	n_{CH_3OH}	$\frac{n_{NH_3}}{n_{CH_3OH}}$	δ_1	δ_2	% (CH_3NH_2)	$\frac{(CH_3NH_2)}{(CH_3)_2NH_4}$
100	50	2	36.8	8.1	19.0%	3.5
100	100	1	59.4	21.8	18.8	1.7
100	200	.5	83.2	47.4	11.9	.75
10	400	.25	95.7	74	4.4	.29



Excess NH_4 will give us better

.29	.37
1376	.37
.36	.15
.22	.09

